

### Assignment No. 4 & 6

#### 1. What is hypothesis testing?

Hypothesis testing is a **statistical method** used to make decisions or inferences about a population based on sample data. It helps determine whether an observed effect is statistically significant or occurred by chance.

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### Key Terms in Hypothesis Testing

1. **Null Hypothesis ( $H_0$ )** – Assumes no effect or no difference.
    - Example: *"There is no difference in average test scores between two groups."*
  2. **Alternative Hypothesis ( $H_1$  or  $H_a$ )** – Assumes there is an effect or difference.
    - Example: *"Group A has a higher average test score than Group B."*
  3. **Significance Level ( $\alpha$ )** – The probability of rejecting  $H_0$  when it is actually true (commonly set at **0.05** or **5%**).
  4. **p-value** – The probability of obtaining results at least as extreme as the observed data, assuming  $H_0$  is true.
    - If **p-value** <  $\alpha$ , reject  $H_0$  (statistically significant).
    - If **p-value**  $\geq \alpha$ , fail to reject  $H_0$  (not significant).
  5. **Type I Error** – Rejecting  $H_0$  when it is actually true (false positive).
  6. **Type II Error** – Failing to reject  $H_0$  when it is actually false (false negative).
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### Steps in Hypothesis Testing

1. **Define  $H_0$  and  $H_1$**  (Formulate hypotheses).

2. **Select a significance level ( $\alpha$ )** (Commonly 0.05).
  3. **Choose the appropriate statistical test** (e.g., t-test, chi-square test).
  4. **Compute the test statistic and p-value** using sample data.
  5. **Make a decision:**
    - If  $p < \alpha$ , reject  $H_0$  (evidence supports  $H_1$ ).
    - If  $p \geq \alpha$ , fail to reject  $H_0$  (no strong evidence for  $H_1$ ).
2. Explain z test, t test (one sampled, two sampled, paired t test) , and ANOVA (one way and two way) and when to perform which test?

### **Z-test:**

- **What it is:** A statistical test used to determine whether two population means are different when the population *standard deviations are known* and the sample size is large (generally  $n \geq 30$ ). It relies on the standard normal distribution (z-distribution).
- **When to use it:**
  - You know the population standard deviation.
  - Your sample size is large ( $n \geq 30$ ).
  - You want to compare the mean of a sample to a known population mean (one-sample z-test) or compare the means of two independent samples (two-sample z-test).
- **Example:** You want to know if the average height of students in a university is different from the national average height, and you know the national standard deviation.

### **T-tests:**

T-tests are used when the population standard deviation is *unknown*, and you have to estimate it from the sample. They use the t-distribution, which accounts for the added uncertainty.

- **One-Sample t-test:**
  - **What it is:** Used to determine if the mean of a single sample is significantly different from a known or hypothesized value.

- **When to use it:** You have one group, and you're comparing its mean to a specific value.
- **Example:** You want to know if the average weight of apples from a particular orchard is different from the standard weight for apples.
- **Two-Sample t-test (Independent):**
  - **What it is:** Used to determine if the means of two *independent* groups are significantly different.
  - **When to use it:** You have two separate groups, and there's no relationship between the individuals in the groups.
  - **Example:** You want to compare the average test scores of students taught by two different methods.
- **Paired t-test (Dependent/Related):**
  - **What it is:** Used to determine if there's a significant difference between the means of two *related* groups (e.g., measurements taken on the same individuals at different times or under different conditions).
  - **When to use it:** You have two groups, and the data points are linked (e.g., before and after measurements on the same subjects).
  - **Example:** You want to see if a weight loss program is effective by comparing the weights of participants before and after the program.

### **ANOVA (Analysis of Variance):**

ANOVA is used to compare the means of *three or more* groups. It tests for differences by analyzing the variance within and between the samples.

- **One-Way ANOVA:**
  - **What it is:** Used to compare the means of three or more independent groups on *one* independent variable (factor).
  - **When to use it:** You have one categorical independent variable with three or more levels (groups), and you want to see if there are significant differences in the means of a continuous dependent variable.
  - **Example:** You want to compare the effectiveness of three different fertilizers on crop yield.
- **Two-Way ANOVA:**

- **What it is:** Used to compare the means of groups when you have *two* independent variables (factors). It can also assess the interaction between these factors.
- **When to use it:** You have two categorical independent variables, each with two or more levels, and you want to see their individual and combined effects on a continuous dependent variable.
- **Example:** You want to study the effect of both fertilizer type (three types) and watering frequency (two levels) on crop yield, and also see if the effect of fertilizer depends on the watering frequency.

3. Explain the steps in hypothesis testing.

Hypothesis testing is a crucial process in statistical inference. It helps us make decisions about populations based on sample data

### **State the Hypotheses:**

- **Null Hypothesis ( $H_0$ ):** This is a statement of no effect or no difference. It's the status quo or what we're trying to disprove. It often includes equality (e.g.,  $\mu = 10$ , there's no difference between groups).
- **Alternative Hypothesis ( $H_1$  or  $H_a$ ):** This is what we're trying to find evidence for. It's the opposite of the null hypothesis. It can be directional (e.g.,  $\mu > 10$ , group A is greater than group B) or non-directional (e.g.,  $\mu \neq 10$ , there's a difference between groups).

### **2. Set the Significance Level (alpha, $\alpha$ ):**

- Alpha represents the probability of rejecting the null hypothesis when it's actually true (Type I error). Common values are 0.05 (5%), 0.01 (1%), or 0.10 (10%). This is the threshold for how unlikely our results must be under the null hypothesis to reject it.

### **3. Choose the Appropriate Test Statistic:**

- Select the statistical test that's appropriate for your data and research question (e.g., t-test, z-test, ANOVA, chi-square). This will depend on factors like sample size, type of data, and the number of groups you're comparing.

### **4. Calculate the Test Statistic:**

- Using your sample data, calculate the value of the chosen test statistic. This measures how far your sample data deviates from what you'd expect under the null hypothesis.
- 5. **Determine the p-value:**
  - The p-value is the probability of obtaining results as extreme as, or more extreme than, the ones observed in your sample *if the null hypothesis is true*. A small p-value means your results are unlikely under the null hypothesis.
- 6. **Make a Decision:**
  - **Compare the p-value to the significance level (alpha):**
    - If the p-value is less than or equal to alpha ( $p \leq \alpha$ ): Reject the null hypothesis. There's statistically significant evidence to support the alternative hypothesis.
    - If the p-value is greater than alpha ( $p > \alpha$ ): Fail to reject the null hypothesis. There's not enough evidence to reject the null hypothesis. This *doesn't* mean the null hypothesis is true; it just means we don't have enough evidence to disprove it.
- 7. **Draw a Conclusion:**
  - State your conclusion in the context of your research question. Don't just say "reject  $H_0$ ." Explain what the results mean in practical terms. For example, "There is sufficient evidence to conclude that the average height of women is different from 5'4"."

### **Important Considerations:**

- **Type I and Type II Errors:**
  - **Type I Error (False Positive):** Rejecting the null hypothesis when it's actually true. The probability of this is alpha.
  - **Type II Error (False Negative):** Failing to reject the null hypothesis when it's actually false. The probability of this is beta ( $\beta$ ).
- **Power:** The power of a test is the probability of correctly rejecting the null hypothesis when it's false ( $1 - \beta$ ). We want tests with high power.
- **Assumptions:** Statistical tests have assumptions that must be met for the results to be valid. Check these assumptions before drawing conclusions.
- **Practical Significance vs. Statistical Significance:** A statistically significant result may not be practically meaningful. Consider the effect size and the real-world implications of your findings.

In the population, the average IQ is 100 with a standard deviation of 15. A team of scientists want to test a new medication to see if it has either a positive or negative effect on intelligence, or not effect at all. A sample of 30 participants who have taken the medication has a mean of 140. Did the medication affect intelligence?

```
import numpy as np
import scipy.stats as stats

mean1 = 100
std = 15
mean2 = 140
size = 30

error = std / np.sqrt(size)

z_score = (mean2 - mean1) / error

alpha = 0.05
critical_value = stats.norm.ppf(1 - alpha / 2)

print(f"Z-score: {z_score}")
print(f"Critical value: ±{critical_value}")

if abs(z_score) > critical_value:
    print("Reject the null hypothesis: The medication has an effect on intelligence.")
else:
    print("Fail to reject the null hypothesis: The medication does not have an effect on intelligence.")
```

### Output:

Z-score: 14.60593486680443

Critical value:  $\pm 1.959963984540054$

Reject the null hypothesis: The medication has an effect on intelligence.

A professor wants to know if her introductory statistics class has a good grasp of basic math. Six students are chosen at random from the class and given a math proficiency test. The professor wants the class to be able to score above 70 on the test. The six students get the following scores: 62, 92, 75, 68, 83, 95. Can the professor have 90% confidence that the mean score for the class on the test would be above 70.

```
import numpy as np
import scipy.stats as stats

scores = [62, 92, 75, 68, 83, 95]

n = len(scores)

sample_mean = np.mean(scores)
sample_std = np.std(scores, ddof=1)

sem = sample_std / np.sqrt(n)

t_statistic = (sample_mean - 70) / sem

df = n - 1

alpha = 0.10
critical_t_value = stats.t.ppf(1 - alpha, df)

print(f"Sample Mean: {sample_mean}")
print(f"Sample Standard Deviation: {sample_std}")
print(f"Standard Error of the Mean: {sem}")
```

```

print(f"T-statistic: {t_statistic}")
print(f"Critical t-value (90% confidence): {critical_t_value}")

if t_statistic > critical_t_value:
    print("Reject the null hypothesis: The class has a good grasp of basic math (mean > 70).")
else:
    print("Fail to reject the null hypothesis: The class may not have a good grasp of basic math (mean <= 70).")

```

### Output:

Sample Mean: 79.16666666666667  
 Sample Standard Deviation: 13.166877635440631  
 Standard Error of the Mean: 5.375355285415509  
 T-statistic: 1.705313636019149  
 Critical t-value (90% confidence): 1.4758840488558216  
 Reject the null hypothesis: The class has a good grasp of basic math (mean > 70).

A clinic provides a program to help their clients lose weight and asks a consumer agency to investigate the effectiveness of the program. The agency takes a sample of 15 people, weighing each person in the sample before the program begins and 3 months later. The results are tabulated below. Determine if the program is effective.

```

import pandas as pd
import scipy.stats as stats

data = {
    "Person": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15],
    "Before": [210, 197, 168, 176, 189, 211, 234, 198, 214, 185, 240, 195, 208, 245, 228],
    "After": [197, 196, 170, 174, 182, 196, 220, 186, 200, 181, 225, 181, 192, 231, 218]
}

```



```

}
df = pd.DataFrame(data)
df["Difference"] = df["Before"] - df["After"]
t_statistic, p_value = stats.ttest_rel(df["Before"], df["After"])
print("Mean difference:", df["Difference"].mean())
print("T-statistic:", t_statistic)
print("P-value:", p_value)

alpha = 0.05
if p_value < alpha:
    print("The program is effective ")
else:
    print("The program is not effective ")

```

### Output:

Mean difference: 9.933333333333334

T-statistic: 6.457368815577421

P-value: 1.5041838796083835e-05

The program is effective

Consider you are performing ML for predicting housing prices you have trained three models and following data summarizes the predicted house price by each model for 5 different trial runs

| Model   | House price predicted (in Lakh Rs.) |         |         |         |         |
|---------|-------------------------------------|---------|---------|---------|---------|
| Code    | Trial 1                             | Trial 2 | Trial 3 | Trial 4 | Trial 5 |
| Model A | 3.5                                 | 3.4     | 3.8     | 3.5     | 3.4     |
| Model B | 3.9                                 | 3.8     | 3.7     | 3.9     | 3.6     |
| Model C | 3.5                                 | 3.3     | 3.6     | 3.5     | 3.8     |

**Program:**

```
import pandas as pd
import matplotlib.pyplot as plt

# Creating the DataFrame
data = {
    "Model": ["Model A", "Model B", "Model C"],
    "Trial 1": [3.5, 3.9, 3.5],
    "Trial 2": [3.4, 3.8, 3.3],
    "Trial 3": [3.8, 3.7, 3.6],
    "Trial 4": [3.5, 3.9, 3.5],
    "Trial 5": [3.4, 3.6, 3.8]
}

df = pd.DataFrame(data)
print("Predicted House Prices Data:")
print(df)

# Plotting the data
plt.figure(figsize=(8, 5))
for i, row in df.iterrows():
```

```
plt.plot(["Trial 1", "Trial 2", "Trial 3", "Trial 4", "Trial 5"], row[1:], marker='o',  
label=row["Model"])
```

```
plt.xlabel("Trials")
```

```
plt.ylabel("Predicted Price (Lakh Rs.)")
```

```
plt.title("House Price Prediction by Different Models")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

**Output:**

Predicted House Prices Data:

|   | Model   | Trial 1 | Trial 2 | Trial 3 | Trial 4 | Trial 5 |
|---|---------|---------|---------|---------|---------|---------|
| 0 | Model A | 3.5     | 3.4     | 3.8     | 3.5     | 3.4     |
| 1 | Model B | 3.9     | 3.8     | 3.7     | 3.9     | 3.6     |
| 2 | Model C | 3.5     | 3.3     | 3.6     | 3.5     | 3.8     |

